WORKED SOLUTION

STEP I 1990, Question 11

Mechanics • Projectiles and Momentum

STEP I 1990, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s1-q11

The key idea

There are three stages: an ordinary projectile up to the top of its flight, an instantaneous collision (the catch), and an instantaneous "explosion" (the push). During the two instantaneous stages gravity has no time to act, so momentum of the shell–Superman system is conserved, separately in each direction. After that the shell is a projectile again, starting from a known height with a known downward speed.

Solution

We model the shell and Superman as particles, ignore air resistance, and take the shell to be fired from, and to land on, horizontal ground. Take x horizontal in the direction of motion and y vertically upwards.

Up to the top of the flight

The shell starts with components

$$u_x = v\cos\tfrac{\pi}{3} = \tfrac{1}{2}v, \qquad u_y = v\sin\tfrac{\pi}{3} = \tfrac{\sqrt{3}}{2}v.$$

At the top the vertical velocity is zero, so the time taken and the height reached are

$$t_1 = \frac{u_y}{g} = \frac{\sqrt{3}\,v}{2g}, \qquad H = \frac{u_y^2}{2g} = \frac{3v^2}{8g}.$$

At this moment the shell is moving horizontally with speed $\frac{1}{2}v$.

The catch

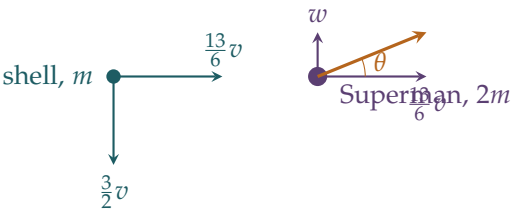
Superman, moving horizontally at $3v$, catches the shell from behind and they move together. Horizontal momentum is conserved during the (instantaneous) catch, so if V

is their common speed,

$$2m(3v) + m\left(\tfrac{1}{2}v\right) = 3mV \quad \Longrightarrow \quad V = \frac{13v}{6}.$$

Neither has any vertical velocity at this instant, so the combined vertical momentum is zero.

The push



Superman now pushes the shell straight down. The push is an internal impulse, so the total momentum of the pair is unchanged in both directions.

Horizontally: the shell keeps horizontal speed $\frac{13}{6}v$. The total horizontal momentum is still $3m \cdot \frac{13}{6}v$, so Superman keeps horizontal speed $\frac{13}{6}v$ as well. (This is no surprise: the impulse is vertical.)

Vertically: the total vertical momentum was zero, so if Superman moves upwards with speed w,

$$2m\,w - m\left(\tfrac{3}{2}v\right) = 0 \quad \Longrightarrow \quad w = \tfrac{3}{4}v.$$

Ask yourself Why can we use conservation of momentum when gravity is acting throughout?

The catch and the push are instantaneous. Gravity is a finite force, so over a time interval of length zero its impulse is zero; only the huge, short-lived internal forces between Superman and the shell matter, and those cancel in pairs.

The rest of the shell's flight

The shell now starts from height $H = \frac{3v^2}{8g}$ with downward vertical speed $\frac{3}{2}v$. Measuring s downwards, it reaches the ground after time t_2 where

$$\tfrac{3}{2}v\,t_2 + \tfrac{1}{2}gt_2^2 = \frac{3v^2}{8g}.$$

Multiplying by $8g$ gives a quadratic in gt_2:

$$4(gt_2)^2 + 12v(gt_2) - 3v^2 = 0 \quad \Longrightarrow \quad gt_2 = \frac{-12v + \sqrt{144v^2 + 48v^2}}{8} = \frac{-12 + 8\sqrt{3}}{8}\,v,$$

taking the positive root since $t_2 > 0$. So

$$t_2 = \left(\sqrt{3} - \tfrac{3}{2}\right)\frac{v}{g}.$$

(This is positive because $\sqrt{3} \approx 1.732 > 1.5$.) The total time of flight is

$$\begin{aligned} T = t_1 + t_2 &= \frac{\sqrt{3}}{2}\frac{v}{g} + \left(\sqrt{3} - \frac{3}{2}\right)\frac{v}{g} \\ &= \frac{3(\sqrt{3}-1)}{2}\frac{v}{g} \;\approx\; 1.10\,\frac{v}{g}. \end{aligned}$$

Superman's direction

Immediately after the push Superman has velocity components $\frac{13}{6}v$ horizontally and $\frac{3}{4}v$ upwards. His trajectory therefore starts at an angle θ above the horizontal with

$$\tan\theta = \frac{\frac{3}{4}v}{\frac{13}{6}v} = \frac{9}{26}, \qquad \theta = \arctan\frac{9}{26} \approx 19.09^\circ,$$

so the angle is 19° to the nearest degree.

Checks

- Without Superman the shell would land at time $2t_1 = \sqrt{3}\,v/g \approx 1.73\,v/g$. Pushing it down should shorten the flight, and indeed $T \approx 1.10\,v/g$ is smaller.
- Mass ratio: the shell is half Superman's mass, so it should get twice his vertical speed in the opposite direction. It does: $\frac{3}{2}v = 2 \times \frac{3}{4}v$.
- Numerically, with $v = 13.7$ and $g = 9.81$, stepping the shell's fall from height H in steps of 10^{-6} gives $t_2 \approx 0.32407\,\text{s}$, matching $(\sqrt{3} - \frac{3}{2})v/g$, and $T \approx 1.5335\,\text{s} = 1.0981\,v/g$.

Common slips

- Forgetting the catch: using the shell's own horizontal speed $\frac{1}{2}v$, or Superman's $3v$, after the push instead of the common speed $\frac{13}{6}v$.
- Giving Superman an upward speed of $\frac{3}{2}v$ by ignoring the masses.
- Taking the negative root of the quadratic, or adding t_2 to $2t_1$ instead of t_1.

Beyond STEP

The catch is a perfectly inelastic collision and the push is its reverse, an explosion: the same conservation law runs both ways, while kinetic energy is lost in one and

supplied in the other. In a first-year course this becomes the general theory of impulsive forces and of a system's centre of mass, whose motion depends only on the external forces, however violently the parts push on one another. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats momentum, impulse and the centre of mass in this spirit.

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